

2014 HKDSE Physics

Paper 1A Multiple Choice Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

Question 1



Two identical scoops of ice cream are taken from a refrigerator and put into paper cup X and vacuum flask Y. At room temperature they melt completely in times t_X and t_Y . Which result and explanation are correct?

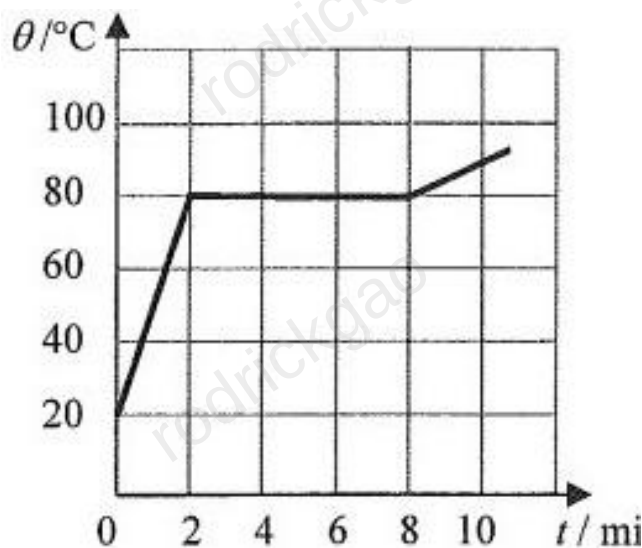
- A. $t_X > t_Y$, as the vacuum flask reduces heat loss to the surroundings.
- B. $t_X > t_Y$, as the vacuum flask retains heat.
- C. $t_Y > t_X$, as the vacuum flask keeps things cold by releasing heat.
- D. $t_Y > t_X$, as the vacuum flask reduces the rate of heat gain from the surroundings.

Official answer: D

The surroundings are warmer than the ice cream, so net heat flows into it. Melting requires latent heat.

Insulation slows heat gain, so the identical scoop in Y takes longer: $t_Y > t_X$. Choose D. A vacuum flask does not actively cool its contents by releasing heat.

Question 2



An insulated solid X is heated at constant power. The graph shows temperature against time. Its specific heat capacity in the solid state is $800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Find its specific latent heat of fusion.

A. 144 kJ kg^{-1}

B. 192 kJ kg^{-1}

C. 202 kJ kg^{-1}

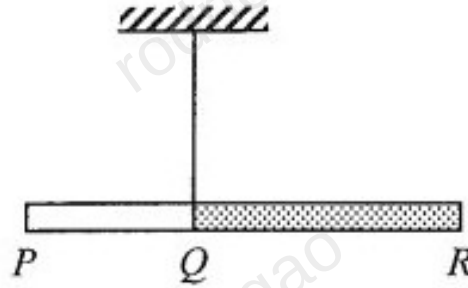
D. It cannot be found because neither the mass nor heater power is given.

Official answer: A

From 0 to 2 min, the solid warms from 20°C to 80°C . Hence $P/m = c\Delta\theta/\Delta t = 800(60)/120 = 400 \text{ W kg}^{-1}$.

The horizontal melting plateau lasts from 2 to 8 min, a duration of 360 s. Thus $l = (P/m)t = 400(360) = 144000 \text{ J kg}^{-1} = 144 \text{ kJ kg}^{-1}$: A. Mass and power cancel in the ratio.

Question 3



A rod PQR of uniform cross-section consists of two materials PQ and QR, each of uniform density. PQ : QR = 2 : 3 in length. Suspended at Q, it stays horizontal. Find the mass ratio of PQ to QR.

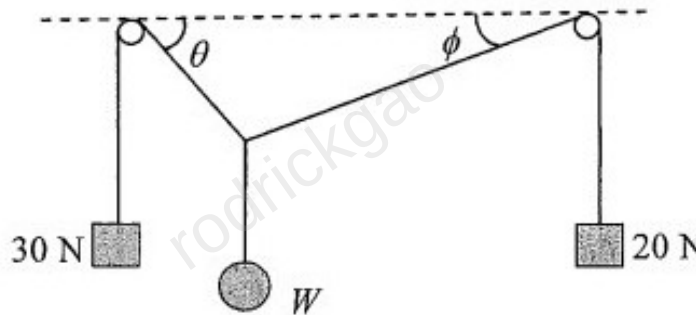
- A. 2 : 3
- B. 1 : 1
- C. 3 : 2
- D. It cannot be found without the density ratio.

Official answer: C

Each uniform section has its centre of gravity at its midpoint. Taking moments about Q gives $m_{PQ}g(L_{PQ}/2) = m_{QR}g(L_{QR}/2)$.

Therefore $m_{PQ}/m_{QR} = L_{QR}/L_{PQ} = 3/2$: C. The shorter arm must carry the larger mass.

Question 4



Two light strings pass over smooth pegs at the same height and support a weight W . Their other ends carry weights 30 N and 20 N. The system is in equilibrium, with the angles θ and ϕ shown. Which deduction is correct?

- A. $W < 50 \text{ N}$.
- B. $W = 50 \text{ N}$.
- C. $W > 50 \text{ N}$.
- D. Nothing can be deduced without knowing θ and ϕ .

Official answer: A

Smooth pegs give tensions of 30 N and 20 N in the respective strings. Only their upward components support W .

Vertical equilibrium gives $W = 30 \sin \theta + 20 \sin \phi$. Both sloping strings have angles less than 90° to the horizontal, so $W < 30 + 20 = 50 \text{ N}$: A. Tension magnitudes cannot simply be added as scalars.

Question 5

A particle moves along a straight line with uniform acceleration. It takes 4 s to travel 36 m, then 2 s to travel the next 36 m. What is its acceleration?

- A. 2.5 m s^{-2}
- B. 3.0 m s^{-2}
- C. 4.0 m s^{-2}
- D. 4.5 m s^{-2}

Official answer: B

For initial velocity u and acceleration a , the first 4 s give $36 = 4u + 8a$. The entire 6 s give $72 = 6u + 18a$.

Subtract 1.5 times the first equation from the second: $18 = 6a$, so $a = 3.0 \text{ m s}^{-2}$; B. The second interval starts with speed $u+4a$, not u .

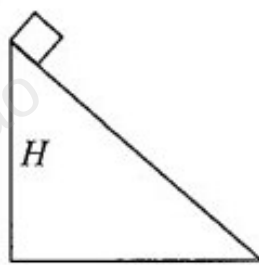
Question 6

Figure (1)

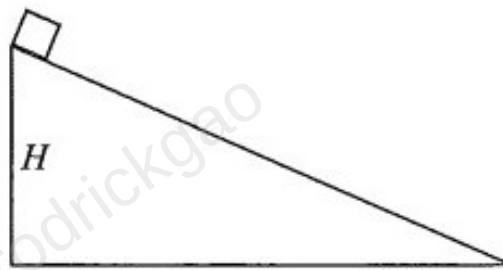


Figure (2)

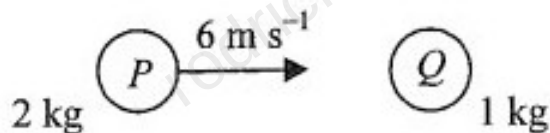
Two identical small blocks slide from rest down smooth inclined planes from the same height H . Their speeds at the bottom are v_1 , v_2 and travel times are t_1 , t_2 . Neglect air resistance. Which is correct?

- A. $v_1 > v_2$ and $t_1 = t_2$
- B. $v_1 > v_2$ and $t_1 < t_2$
- C. $v_1 = v_2$ and $t_1 = t_2$
- D. $v_1 = v_2$ and $t_1 < t_2$

Official answer: D

Conservation of mechanical energy gives $mgH = \frac{1}{2}mv^2$, so both bottom speeds are $\sqrt{2gH}$.

Along a plane inclined at θ , $a = g \sin \theta$ and $t = v/a$. Plane (1) is steeper, so its acceleration is larger and its time shorter. Therefore $v_1 = v_2$, $t_1 < t_2$: D.

Question 7

Sphere P of mass 2 kg collides head-on with sphere Q of mass 1 kg, initially at rest. P has speed 6 m s^{-1} just before collision. Both move in the same direction after collision. Which could be the speed(s) of Q just after collision?

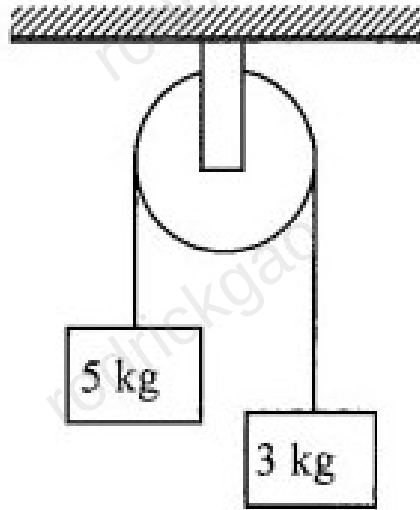
- (1) 2 m s^{-1}
- (2) 4 m s^{-1}
- (3) 6 m s^{-1}

- A. (1) only
- B. (1) and (2) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: C

Momentum conservation gives $12 = 2v_P + v_Q$. For $v_Q = 2, 4, 6 \text{ m s}^{-1}$, the corresponding v_P are 5, 4, 3 m s^{-1} .

At the end of collision Q must not be slower than P behind it, otherwise they would still be approaching. The 2 m s^{-1} case fails this condition. The 4 and 6 m s^{-1} cases permit common motion or separation, and both reduce kinetic energy. Choose C.

Question 8

Blocks of masses 5 kg and 3 kg are joined by a light string over a frictionless fixed light pulley. Find their acceleration magnitude when released, in terms of g . Neglect air resistance.

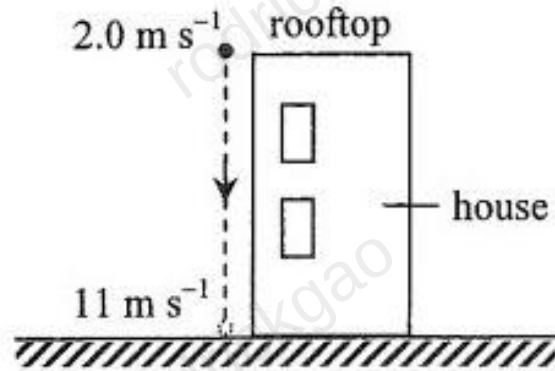
- A. g
- B. $g/2$
- C. $g/4$
- D. $g/8$

Official answer: C

For the 5 kg block moving down: $5g - T = 5a$. For the 3 kg block moving up: $T - 3g = 3a$. The ideal pulley and string give equal tension and acceleration magnitude.

Add the equations: $(5-3)g = (5+3)a$, hence $a = g/4$: C. The driving force is the weight difference, but the accelerated mass is the sum.

Question 9



A particle is projected vertically downwards from a rooftop at 2.0 m s^{-1} and reaches the ground at 11 m s^{-1} . Estimate the height. Neglect air resistance; $g = 9.81 \text{ m s}^{-2}$.

A. 3.3 m

B. 6.0 m

C. 6.5 m

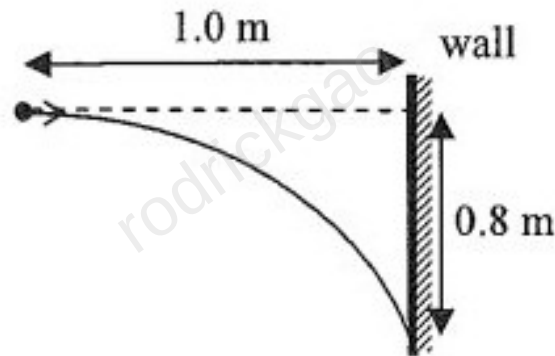
D. 12 m

Official answer: B

Take downward as positive and apply $v^2 = u^2 + 2gh$.

$h = (11^2 - 2.0^2) / (2 \times 9.81) = 5.96 \text{ m} \approx 6.0 \text{ m}$: B. Subtract the initial speed squared, not the speed itself.

Question 10



* A particle is projected horizontally towards a vertical wall 1.0 m away. It hits the wall 0.8 m below its projection point. Find its initial speed. Neglect air resistance; $g = 9.81 \text{ m s}^{-2}$.

A. 2.0 m s^{-1}

B. 2.5 m s^{-1}

C. 5.0 m s^{-1}

D. 6.3 m s^{-1}

Official answer: B

The initial vertical velocity is zero. Vertical motion gives $0.8 = \frac{1}{2}gt^2$, hence $t = \sqrt{(1.6/9.81)} = 0.404 \text{ s}$.

Horizontal velocity stays constant, so $u = 1.0/t = 2.48 \text{ m s}^{-1} \approx 2.5 \text{ m s}^{-1}$: B. Use the vertical drop to obtain the flight time first.

Question 11

* An astronaut in a spacecraft moving in a circular orbit around the Earth is apparently weightless because

- A. the astronaut is too far away to feel the Earth's gravitational force.
- B. the astronaut and spacecraft have the same acceleration towards the Earth.
- C. the Earth's gravitational force on the astronaut is balanced by the floor reaction.
- D. the Earth's gravitational force on the astronaut is balanced by the centripetal force.

Official answer: B

The astronaut and spacecraft are both in free fall, accelerating towards the Earth together. No supporting normal force is needed, so the apparent weight is zero: B.

Gravity is still present and supplies the centripetal force. Centripetal force is the name of the inward resultant, not an extra force balancing gravity.

Question 12

* A satellite moves in a circular orbit at a height above the Earth equal to the Earth's radius. Find its acceleration in terms of the surface gravitational acceleration g .

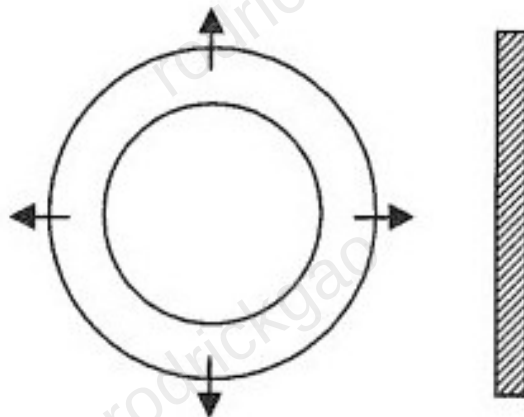
- A. $g/8$
- B. $g/4$
- C. $g/2$
- D. g

Official answer: B

Let the Earth's radius be R . The orbital radius measured from the centre is $R+R = 2R$, not R .

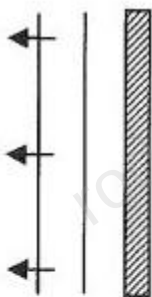
$a = GM/(2R)^2 = \frac{1}{4}GM/R^2 = g/4$: B. Gravitational acceleration varies inversely as the square of distance from the centre.

Question 13



Two circular pulses produced by water drops in a ripple tank are reflected by a straight barrier. Which diagram best shows the reflected pulses?

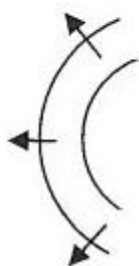
A.



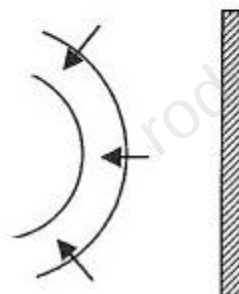
B.



C.



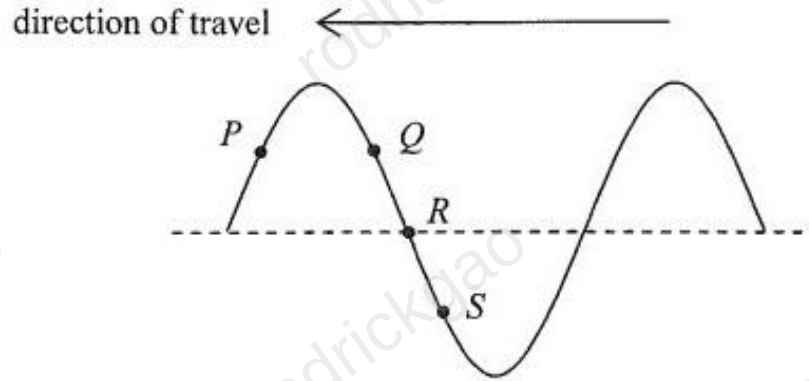
D.

**Official answer: B**

The reflected circular wavefronts behave as if emitted by the mirror image of the source behind the barrier. They move away from that virtual source and back into the water.

Only B has diverging arcs whose centre lies behind the barrier. A incorrectly makes them plane; C places the effective centre on the water side; D shows convergence instead of divergence.

Question 14



A transverse wave travels left along a long string. Which statements correctly describe the motions of particles P, Q, R and S at the instant shown?

- (1) P is moving upwards.
- (2) Q and S are moving in opposite directions.
- (3) R is momentarily at rest.

A. (1) only

B. (3) only

C. (1) and (2) only

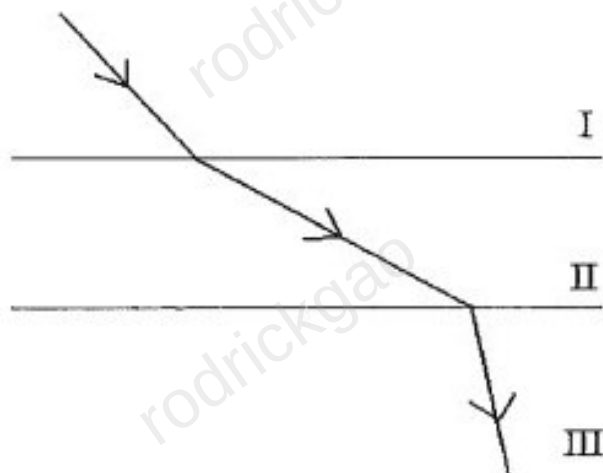
D. (2) and (3) only

Official answer: A

For a left-moving wave $y = f(x+vt)$, particle velocity has the same sign as the local spatial slope. P is on a rising slope, so it moves up. Q and S are both on a falling slope, so both move down.

R crosses the equilibrium line on a steep slope and is not at rest. For this sinusoidal travelling wave, rest occurs at crests and troughs, not zero displacement. Only (1) is correct: A.

Question 15



The figure shows a light ray travelling from medium I to medium III through parallel boundaries. Arrange the light speeds in ascending order.

- A. $I < III < II$
- B. $II < III < I$
- C. $III < I < II$
- D. $III < II < I$

Official answer: C

Measure each angle from the vertical normal. The diagram gives $\theta_{III} < \theta_I < \theta_{II}$. Parallel boundaries make all normals parallel.

Snell's law gives $n \sin \theta$ constant. Since $v = c/n$, v is proportional to $\sin \theta$ along this ray. Thus $v_{III} < v_I < v_{II}$: C.

Question 16

Plane sea waves pass through a gap in a sea wall and diffract. Assuming frequency remains unchanged, which changes increase diffraction?

- (1) The gap becomes narrower.
- (2) The wavelength increases.
- (3) The amplitude increases.

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

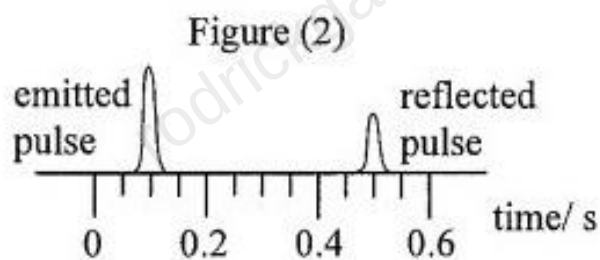
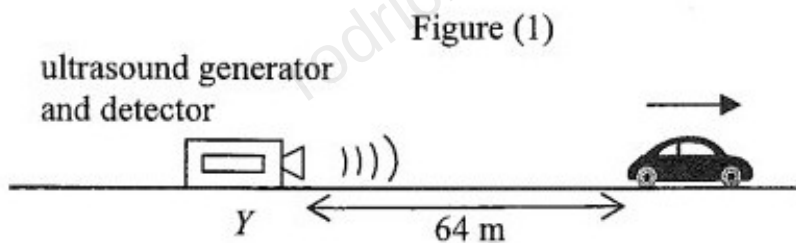
D. (1), (2) and (3)

Official answer: A

Diffraction becomes more pronounced when the wavelength-to-gap-width ratio λ/a increases. A narrower gap or a longer wavelength increases this ratio.

Amplitude changes wave energy, not the diffraction angle in this model. Only (1) and (2) are correct: A. Fixed frequency does not by itself fix wavelength if wave speed changes.

Question 17



A car moves at uniform speed away from a stationary ultrasound generator and detector at Y. A pulse is emitted when the car is 64 m from Y. The emitted and reflected pulses appear on the CRO as shown. Estimate the car speed. The ultrasound speed in air is 340 m s^{-1} .

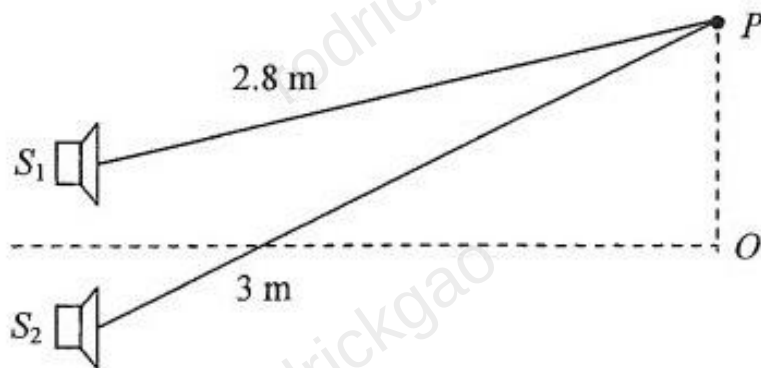
- A. 16 m s^{-1}
- B. 20 m s^{-1}
- C. 24 m s^{-1}
- D. 32 m s^{-1}

Official answer: B

The pulses are at 0.10 s and 0.50 s, so the round-trip time is 0.40 s. Both legs cover the same distance between Y and the reflection point at the same sound speed, so reflection occurs 0.20 s after emission.

The reflection point is $340(0.20) = 68 \text{ m}$ from Y. During those 0.20 s the car travels $68 - 64 = 4 \text{ m}$, giving $v = 4/0.20 = 20 \text{ m s}^{-1}$: B. Do not use 0.50 s as the elapsed time.

Question 18



Loudspeakers S_1 and S_2 are connected to a signal generator but produce sound in anti-phase. O is equidistant from them. $S_1P = 2.8$ m and $S_2P = 3.0$ m. With wavelength 10 cm, what interference occurs at O and P respectively?

- A. Destructive; constructive
- B. Constructive; constructive
- C. Destructive; destructive
- D. Constructive; destructive

Official answer: C

The sources start π radians out of phase. At O the path difference is zero, so the waves remain in anti-phase and interfere destructively.

At P , $\Delta r = 3.0 - 2.8 = 0.20$ m $= 2\lambda$. This adds 4π phase difference, leaving the original anti-phase relation unchanged. P is also destructive: C. Integer-wavelength path difference only means constructive interference for in-phase sources.

Question 19

Which statements about sound waves are correct?

- (1) Sound waves are electromagnetic waves.
- (2) Sound waves cannot travel in a vacuum.
- (3) Sound waves cannot form stationary waves.

A. (2) only

B. (3) only

C. (1) and (2) only

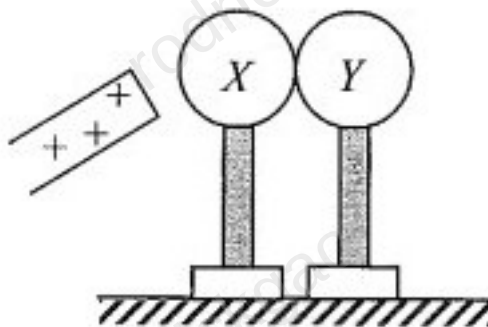
D. (1) and (3) only

Official answer: A

Sound is a mechanical disturbance requiring a material medium, not an electromagnetic wave. Thus (1) is false and (2) is true.

Incident and reflected sound of the same frequency can form stationary waves, as in a resonating air column. Statement (3) is false, so choose A.

Question 20



Two insulated uncharged metal spheres X and Y touch. A positive rod is brought near X. X is touched momentarily by a finger, then Y is moved away to separate the spheres. The rod is removed afterwards. What are the final charges on X and Y respectively?

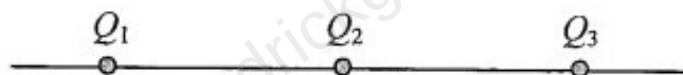
- A. Uncharged; uncharged
- B. Uncharged; positive
- C. Negative; uncharged
- D. Negative; negative

Official answer: C

The positive rod attracts electrons towards X. Momentary earthing supplies electrons from the Earth, neutralising the electron deficit on the farther side.

In the induction model used by this question, the excess electrons remain attracted near X while Y is separated with the rod still present. X retains negative charge and Y is uncharged: C. The order matters: removing the rod before separation would allow redistribution.

Question 21



Three point charges Q_1 , Q_2 , Q_3 lie on a straight line with Q_2 at the midpoint. The resultant electrostatic force on each is zero. Which signed charges, in the same arbitrary units, are possible? Options list Q_1 , Q_2 , Q_3 .

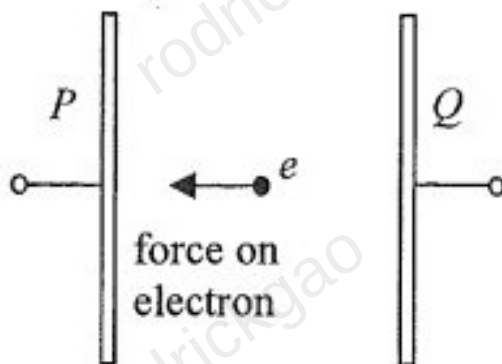
- A. +2, +1, +2
- B. +2, -1, +2
- C. -4, +1, +4
- D. -4, +1, -4

Official answer: D

Zero force on the middle charge requires the end charges to have the same sign and magnitude. To balance an end charge, the middle charge must have the opposite sign.

Let adjacent separation be d . At Q_1 , $k|Q_1Q_2|/d^2 = k|Q_1Q_3|/(2d)^2$, so $|Q_3| = 4|Q_2|$. The set -4, +1, -4 meets all conditions: D.

Question 22



Parallel metal plates P and Q are maintained at a potential difference by a battery. An electron between them experiences force 8.0×10^{-18} N towards P. Which describes the electric field?

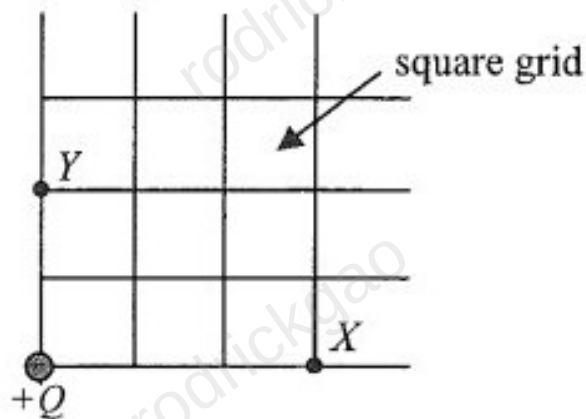
- A. 0.02 N C^{-1} from Q to P
- B. 0.02 N C^{-1} from P to Q
- C. 50 N C^{-1} from Q to P
- D. 50 N C^{-1} from P to Q

Official answer: D

The field magnitude is $E = F/e = (8.0 \times 10^{-18}) / (1.60 \times 10^{-19}) = 50 \text{ N C}^{-1}$.

An electron has negative charge, so its force is opposite to the field. Force is towards P, hence E points from P to Q: D.

Question 23



* An isolated point charge $+Q$ is shown on a square grid. If the electric potential at X is V , what is the potential at Y?

A. $(2/3)V$

B. $(3/2)V$

C. $(4/9)V$

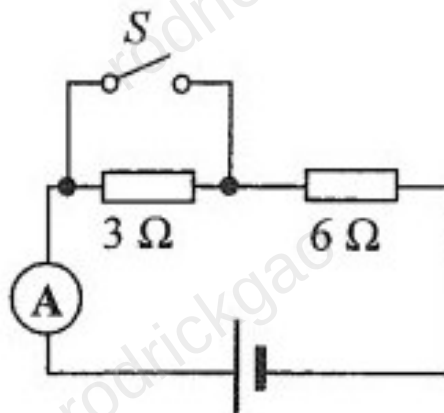
D. $(9/4)V$

Official answer: B

From the grid, X is three cell widths from $+Q$ and Y is two cell widths away. Point-charge potential relative to infinity is kQ/r .

$V_Y/V_X = r_X/r_Y = 3/2$, so $V_Y = (3/2)V$: B. Potential follows $1/r$, not the inverse-square field law.

Question 24



The cell has constant e.m.f. and fixed internal resistance. Closing S gives an ammeter reading of 3.0 A. Which reading is possible with S open?

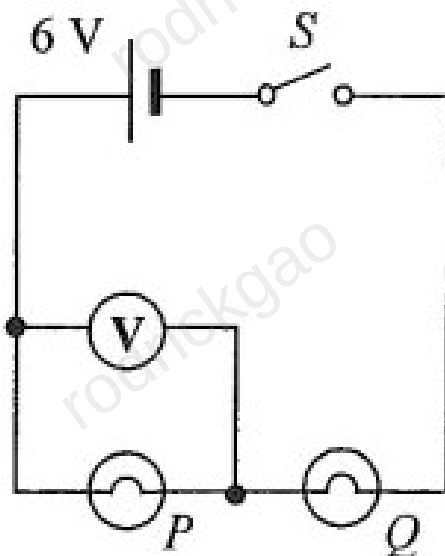
- A. 1.6 A
- B. 2.0 A
- C. 2.4 A
- D. 3.2 A

Official answer: C

Closing S bypasses $3\ \Omega$, so $\varepsilon = 3(6+r)$. Opening S makes the external resistance $9\ \Omega$ and $I = \varepsilon/(9+r) = 3(6+r)/(9+r)$.

For positive internal resistance r , $2\ \text{A} < I < 3\ \text{A}$. Only 2.4 A is possible (it corresponds to $r = 6\ \Omega$): C. The 2.0 A result would require zero internal resistance.

Question 25



Bulbs P and Q are connected to a 6 V cell of negligible internal resistance. The voltmeter reads 6 V when S is closed. Which is possible?

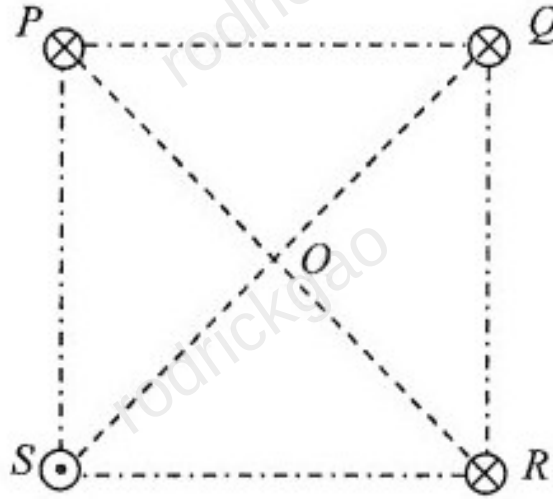
- A. Both P and Q are short-circuited.
- B. Both P and Q are burnt out and become open circuit.
- C. P is short-circuited or Q is burnt out and becomes open circuit.
- D. P is burnt out and becomes open circuit or Q is short-circuited.

Official answer: D

The voltmeter measures the p.d. across P. If P is open circuit and Q is intact, negligible current flows and the p.d. across Q is negligible; the full 6 V appears across P.

If Q is short-circuited and P is intact, P is effectively connected directly across the cell and again has 6 V. These are the two alternatives in D. A short across P would instead give a zero reading.

Question 26



Four long parallel wires at the vertices P, Q, R and S of a square carry equal currents. P, Q and R carry current into the paper; S carries current out. The resultant magnetic field at centre O points along

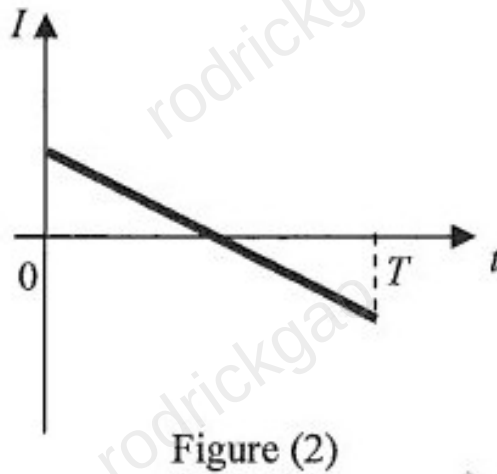
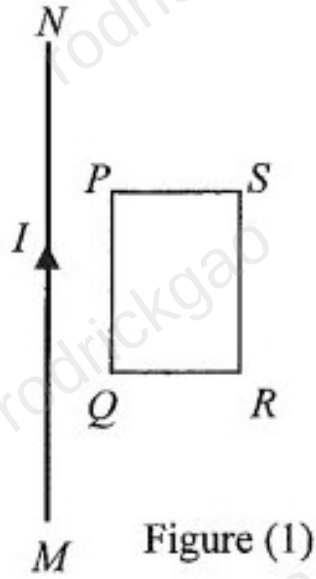
- A. OP
- B. OQ
- C. OR
- D. OS

Official answer: A

All wires are equally distant from O and produce equal field magnitudes there. The fields from opposite wires P and R cancel because their currents have the same direction.

By the right-hand grip rule, Q (current into the paper) gives a field towards P at O. S (current out) also gives a field towards P. They add along OP: A.

Question 27



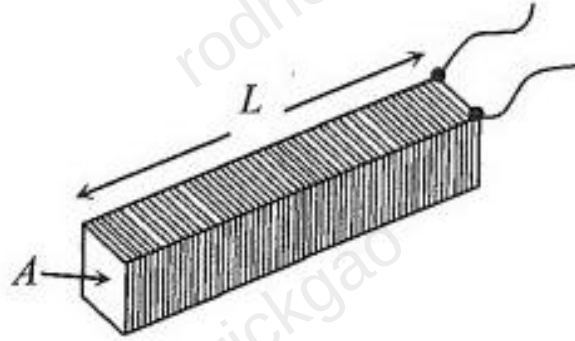
A long straight wire MN and rectangular coil PQRS are fixed in the same plane. Current I is positive from M to N and varies with time as shown. What is the induced-current direction in the coil during 0 to T?

- A. First anticlockwise, then clockwise.
- B. First clockwise, then anticlockwise.
- C. Anticlockwise throughout.
- D. Clockwise throughout.

Official answer: D

Positive upward wire current produces flux into the paper through the coil. Before I reaches zero that inward flux decreases, so the induced current produces inward flux to oppose the decrease.

After I becomes negative, outward flux increases; the induced current still produces inward flux to oppose that increase. Thus it is clockwise throughout: D. Its direction depends on the change in flux, not the sign of I .

Question 28

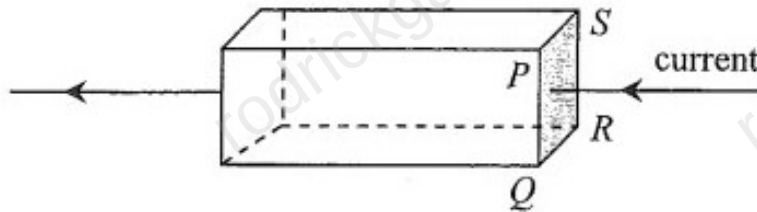
A closely packed long solenoid has cross-sectional area A , length L and N turns, carrying a constant direct current. Which change increases the magnetic flux density at its central cross-section? Options list length, area and total turns.

- A. $2L$, $2A$, $2N$
- B. L , $2A$, N
- C. $2L$, A , N
- D. L , A , $2N$

Official answer: D

For a long solenoid, $B = \mu_0(N/L)I$. With current and core material unchanged, only turns per unit length N/L determine B in this approximation.

A leaves N/L unchanged; B changes only area; C halves N/L ; D doubles N/L . Therefore only D increases B . Do not confuse flux density B with flux BA .

Question 29

* Current flows from right to left in a metal block with cross-section PQRS. With a uniform magnetic field applied, side PQ has higher electric potential than side SR. Which field direction is possible?

- A. P to Q
- B. Q to P
- C. P to S
- D. S to P

Official answer: B

Conduction electrons drift right, opposite to conventional current. Higher potential on PQ means electron accumulation on the opposite side SR, leaving PQ relatively positive.

An upward field Q to P gives $\mathbf{v} \times \mathbf{B}$ towards the front PQ for a positive charge moving right; multiplying by the negative electron charge reverses the force towards SR. This produces the stated polarity: B.

Question 30

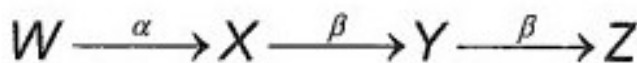
* A heater connected to 10 V d.c. dissipates power P . With a sinusoidal a.c. supply it dissipates $P/2$. Find the a.c. r.m.s. voltage, assuming constant heater resistance.

- A. 5 V
- B. $5\sqrt{2}$ V
- C. 10 V
- D. $10\sqrt{2}$ V

Official answer: B

For a resistor, d.c. power is $P = 10^2/R$. The average a.c. power is V_{rms}^2/R , so $V_{\text{rms}}^2/R = \frac{1}{2}(100/R)$.

$V_{\text{rms}} = \sqrt{50} = 5\sqrt{2}$ V: B. This is already the r.m.s. value; do not divide by $\sqrt{2}$ a second time.

Question 31

Nucleus W decays to Z through $W \rightarrow X$ (α), $X \rightarrow Y$ (β), $Y \rightarrow Z$ (β). Which statements are correct?

- (1) X has one more proton than Y .
- (2) W has two more neutrons than X .
- (3) W and Z are isotopes of the same element.

- A. (1) only
- B. (2) only
- C. (1) and (3) only

D. (2) and (3) only

Official answer: D

Here β denotes β -minus decay: a neutron becomes a proton, so Y has one more proton than X , making (1) false. Alpha decay removes two protons and two neutrons, so W has two more neutrons than X : (2) is true.

Across the whole chain, one alpha decay reduces proton number by 2 and two beta-minus decays restore it by 2. The final mass number is four lower. W and Z have equal proton numbers but different mass numbers, so (3) is true. Choose D.

Question 32

A GM counter close to a source emitting α and γ radiation records 450 counts per minute; background is 50 counts per minute. Materials placed between source and counter give the following readings. Which set of x, y, z is most suitable?

Material	Recorded counts per minute
None	450
Cardboard	x
1 mm aluminium	y
2 mm lead	z

A. 300, 300, 100

B. 300, 100, 50

C. 100, 100, 0

D. 100, 50, 50

Official answer: A

Cardboard stops alpha radiation. A 1 mm aluminium sheet also stops alpha but has little effect on gamma in this school-level comparison, so x and y should be approximately equal.

Lead reduces gamma more strongly, but 2 mm does not completely remove it. The recorded z should remain above the 50-count background. The best set is 300, 300, 100: A. These are measured counts including background, not background-subtracted rates.

Question 33

* A radium nucleus decays to a radon nucleus and an α -particle, releasing 4.9 MeV. Compared with the mass of the radium nucleus, the total mass of the radon nucleus and α -particle is

A. 5.4×10^{-11} kg less.

B. 5.4×10^{-11} kg more.

C. 8.7×10^{-30} kg less.

D. 8.7×10^{-30} kg more.

Official answer: C

Convert the released energy: $E = 4.9 \times 10^6 \times 1.60 \times 10^{-19} = 7.84 \times 10^{-13}$ J. Mass defect is $\Delta m = E/c^2 = (7.84 \times 10^{-13}) / (3.00 \times 10^8)^2 = 8.71 \times 10^{-30}$ kg.

Released kinetic energy comes from a decrease in total rest mass. The products therefore have 8.7×10^{-30} kg less total rest mass: C. Convert MeV to joules before using $E = \Delta mc^2$.